



Lecture 02: Research Methods in Psychology

“Asking Questions, Getting Answers”

1. What is Research in Psychology?

Definition

Psychological research is the **systematic, scientific study of behavior and mental processes**.

It uses **empirical methods** (observation, measurement, and experimentation) to understand how people think, feel, and act.

Psychological research aims to **develop theories, test hypotheses, and apply findings** to real-life problems such as mental health, education, workplace behavior, and social issues.

Objectives of Psychological Research

Goal	Description	Example
Description	To observe and describe behavior as it occurs naturally.	Observing how children interact in a playground.
Explanation	To determine why behavior occurs.	Studying whether aggression in children is caused by frustration.
Prediction	To predict how people will behave in certain situations.	Predicting that sleep deprivation leads to lower academic performance.
Control	To influence or modify behavior.	Using therapy to reduce anxiety.

Importance of Research in Psychology

1. **Evidence-based Understanding:** Replaces myths and opinions with tested knowledge.
 2. **Improves Human Welfare:** Research findings improve education, health, business, and social policy.
 3. **Theory Development:** Theories are built, tested, refined, or rejected through research.
 4. **Scientific Progress:** Leads to innovation in therapy, learning methods, and diagnostics.
 5. **Informs Practice:** Guides professionals like counselors, teachers, and clinicians.
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Example:

Early research on **classical conditioning (Pavlov)** and **operant conditioning (Skinner)** laid the foundation for behavioral therapies still used today.

2. What is Critical Thinking?

Definition:

Critical thinking is the **intentional process of analyzing and evaluating information objectively** in order to make reasoned and evidence-based judgments.

It means not accepting statements at face value — instead, questioning assumptions, testing evidence, and considering alternative explanations.

Stephen D. Brookfield's (1987) Components of Critical Thinking:

1. **Identifying and Challenging Assumptions**
 - Recognize hidden biases, values, and beliefs behind arguments.

- Example: “Men are naturally better at math” — a critical thinker questions whether this stereotype has scientific support.

2. Recognizing the Importance of Context

- Behavior and beliefs vary across situations and cultures.
- Example: Eye contact may show respect in one culture but rudeness in another.

3. Imagining and Exploring Alternatives

- Considering multiple explanations or solutions.
- Example: A student scoring poorly might be unmotivated — or anxious, tired, or facing personal challenges.

4. Reflective Skepticism

- Withholding belief until sufficient evidence is available.
- Example: Questioning viral claims like “Listening to Mozart makes babies smarter.”

Why Critical Thinking is Essential in Psychology

- Prevents blind acceptance of “common sense” or “folk psychology.”
- Encourages evidence-based reasoning.
- Helps evaluate research validity and reliability.
- Improves decision-making and ethical reasoning.
- Encourages **scientific skepticism**, not cynicism — open-minded but cautious thinking.

3. What Makes Something Scientific?

A study is **scientific** when it adheres to the **scientific method**, ensuring reliability, objectivity, and empirical testing.

Core Principles of Scientific Inquiry

1. Empiricism:

Knowledge must come from **observation and evidence**, not opinion.

2. **Falsifiability:**

A theory must be **testable and disprovable** (Popper, 1959).

Example: “All swans are white” is falsifiable — one black swan disproves it.

3. **Replication:**

Findings must be **repeatable** by other researchers.

4. **Objectivity:**

Personal bias must be minimized through standardized procedures.

5. **Systematic Observation:**

Data must be gathered in a structured, methodical way.

6. **Quantifiability:**

Concepts should be measurable (using operational definitions).

7. **Probabilistic Reasoning:**

Psychological research deals with probabilities, not certainties — human behavior varies across contexts.



Example: Wundt’s First Psychological Experiment

Wilhelm Wundt (1832–1920), the “father of experimental psychology,” studied **reaction time** — how long it takes a person to respond to a stimulus.

He found that making a decision (red or green light) took longer than a simple reflex action — showing that mental processes could be **measured scientifically**.



4. Steps in the Research Process

1. Identify a Problem / Research Question

- Example: “Does social media use cause anxiety in teenagers?”

2. Review the Literature

- Study previous research to identify gaps and avoid repetition.

3. Formulate a Hypothesis

- A **testable statement** predicting the relationship between variables.
- Example: “Increased social media use leads to higher anxiety levels.”

4. **Select Research Method**

- Choose design: experimental, correlational, survey, case study, etc.

5. **Collect Data**

- Use valid tools: tests, interviews, surveys, observations.

6. **Analyze Results**

- Apply statistical or qualitative analysis to test hypotheses.

7. **Report Findings**

- Share results through journals, conferences, or reports.

8. **Replication**

- Others repeat the study to verify results.

5. Types of Research in Psychology

A.

Experimental (Causal) Research

Purpose:

To establish **cause-and-effect** relationships.

Structure:

- **Independent Variable (IV):** Manipulated factor.
- **Dependent Variable (DV):** Measured outcome.
- **Control Group:** No treatment.
- **Experimental Group:** Receives treatment.

Example:

Testing whether caffeine (IV) improves memory (DV).

Control: Participants receive no caffeine.

Strengths:

- Establishes causality.
- High control and internal validity.

Limitations:

- Artificial settings reduce realism.
 - Not always ethical (e.g., studying trauma).
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B.

Non-Experimental Research

Definition:

No variable manipulation — researcher observes naturally occurring behavior.

Purpose:

Used when experiments are impossible or unethical.

Examples:

- Studying effects of parental divorce on children.
- Analyzing historical data (e.g., post-war mental health trends).

Strengths:

- Real-world context.
- Can explore phenomena over long periods.

Limitations:

- No control over variables.
 - Cannot infer causality.
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C.**Descriptive Research****Goal:**

To describe characteristics or behaviors in detail.

Methods:**1. Case Study:**

- Deep analysis of a single individual or group.
- Example: Freud's patient studies (e.g., "Little Hans").

2. Naturalistic Observation:

- Observing behavior in its natural environment.
- Example: Watching playground interactions.

3. Survey Research:

- Using questionnaires to collect large-scale data.

Strengths:

- Provides detailed descriptions.

Weakness:

- Cannot explain causes.
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D.**Correlational Research**

Goal:

To measure the **relationship between variables**.

Example:

Studying the relationship between stress and academic performance.

Types of Correlation:

- **Positive:** Both variables increase (e.g., height and weight).
- **Negative:** One increases, the other decreases (e.g., stress and sleep).
- **Zero:** No relationship.

Important:

Correlation does not imply causation.

Two variables may be related because of a third variable (confounding variable).

E.**Quasi-Experimental Research****Definition:**

Similar to experiments but without random assignment.

Example: Comparing two existing school classes taught with different methods.

Used When:

Ethical or practical issues prevent random assignment.

F.**Longitudinal vs Cross-Sectional Studies**

Type	Description	Example	Advantage	Limitation
Longitudinal	Studies same group over years	Tracking memory decline in aging adults	Shows changes over time	Time-consuming & expensive
Cross-sectional	Compares different groups at one time	Comparing teens & adults on sleep habits	Quick & inexpensive	No cause-effect or time analysis

6. Problems and Biases in Research

1. Sampling Bias:

- Participants don't represent population.
- Example: Only college students used → results not generalizable.

2. Experimenter Bias:

- Researcher's expectations subtly influence outcomes.

3. Demand Characteristics:

- Participants act how they think they should.

4. Confirmation Bias:

- Researcher interprets data to fit their hypothesis.

5. Placebo Effect:

- Participants improve because they believe they received treatment.

6. Ethical Concerns:

- Must protect participants:
 - Informed consent
 - Confidentiality
 - Right to withdraw
 - No physical or psychological harm
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7. Quantitative vs Qualitative Research

Feature	Quantitative Research	Qualitative Research
Focus	Numbers and measurements	Meanings and experiences
Goal	Test hypotheses, generalize results	Explore depth, understand context
Approach	Objective, structured	Subjective, open-ended
Data Type	Numerical (scores, scales)	Textual (words, images)
Analysis	Statistical	Thematic / narrative
Example	Survey on stress levels	Interview on coping experiences
Strength	Reliable, generalizable	Deep, detailed insights
Weakness	May ignore context	Hard to generalize

Quantitative Methods Include:

- Experiments
- Surveys
- Questionnaires
- Database analysis
- Statistical modeling

Qualitative Methods Include:

- In-depth interviews
- Focus group discussions
- Participant observations
- Case studies
- Document analysis

8. Folk Psychology vs Scientific Psychology

Folk Psychology:

Refers to everyday beliefs and “common sense” ideas people have about behavior and the mind.

These are often influenced by culture, experience, and intuition rather than science.

Scientific Psychology:

Uses the scientific method to test ideas with evidence and reject unsupported claims.

✗ Common Myths of Folk Psychology

Claim	Scientific Finding
Opposites attract	Similarity predicts attraction.
Familiarity breeds contempt	Familiarity usually increases liking.
More people = more help	Bystander effect: people are less likely to act.
Visual vs verbal learners	Learning styles theory lacks evidence.
Hypnosis is fake	Has real psychological effects in pain reduction and therapy.
Subliminal ads work	No strong evidence.
Mozart makes babies smarter	Only short-term effects.
Old age = unhappiness	Older adults often report higher life satisfaction.
Trust your first answer	Changing answers can improve accuracy.
Stress causes ulcers	Caused by bacteria (H. pylori).
Positive thinking cures cancer	No direct causal link.
Parenting solely determines personality	Genetics and peers also shape traits.
Inkblot tests reveal personality	Low reliability and validity.
Interviews predict job success	Structured tests are better predictors.

9. Cultural Reflection: Psychology Beyond the West

The lecture ends by reminding us that most “folk psychology” ideas come from the **Western world**.

Students are encouraged to **reflect on local cultural beliefs** — for example:

- “Left-handed people are unlucky.”
- “Dreams always predict the future.”
- “Mental illness is caused by evil spirits.”

Psychology encourages evaluating such beliefs **scientifically**, not dismissively — combining **critical thinking** with **cultural awareness**.



FINAL SUMMARY TABLE (Exam Quick Revision)

Topic	Core Idea	Key Example
Critical Thinking	Evaluate evidence & assumptions	Questioning myths about mental illness
Scientific Method	Systematic, testable, falsifiable	Wundt’s reaction time experiment
Experimental Research	Manipulate variables to find cause	Effect of caffeine on memory
Non-Experimental	Observe natural situations	Studying stress during exams
Descriptive	Describe behavior	Case study on Phineas Gage
Correlational	Measure relationships	Stress vs GPA
Quantitative	Numerical data, statistical	Survey data analysis
Qualitative	Narrative data, meanings	Interview about anxiety
Folk Psychology	Common myths	“Opposites attract”
Scientific Psychology	Evidence-based	Similarity leads to liking

